Pray Somaldo

+62 82128660704 | praysomaldo95@gmail.com | github.com/praysmld | prays.site

Education

Master of Computer Science — Indonesia
University of Indonesia (one of the highest ranking university in Indonesia) — *Feb. 2019 – Jan 2021*

Experience

Lead AI Engineer — Jan. 2026 – Now
Simplify — *Jakarta, Indonesia*

- Led a team of 15 AI engineers to develop AI agent to simplify software development life cycle.
- Managed expectations between stakeholders and product owners about AI agents.
  Technology Used: Langchain, langgraph, LLM, Multi-modal

AI Technical Lead — Nov. 2025 – Jan. 2026
PT SMART Tbk — *Jakarta, Indonesia*

- Led a team of 2 AI engineers and 2 software engineers to develop a computer vision-based palm fruit quality assessment project.
- Refactored loose fruit detection codebase, reducing development time by 30% and automating deployment pipelines.
  Technology Used: Computer Vision, Python, deployment automation

AI Advisor (Part-time) — Nov. 2025 – Present
Links — *United States (remote)*

- Advised engineering teams on AI prompting strategies, LLM selection criteria, and managing client expectations for AI capabilities.
- Contributed to building AI agent systems with MCP (Model Context Protocol) server integration for enhanced interoperability.
  Technology Used: LLM, AI Agents, MCP servers, prompt engineering

Senior AI Engineer — July 2024 – June 2025
Polyrific — *New York, US. remote from Indonesia*

- Integrated multiple large language model (LLM) APIs—including OpenAI, Groq, Mistral, Anthropic, and Amazon Bedrock—with third-party platforms such as Gmail, Jira, and Slack to enhance AI-driven workflows and reduce latency by 20%.
- Conducted research on applying knowledge graphs and semantic search to build knowledge bases for LLM applications.
- Built a multi-modal conversational chatbot capable of generating podcast-style audio content and images by leveraging Azure text-to-speech AI services and Flux, extending user interaction beyond text.
- Designed agentic AI systems to enhance chatbot memory: one for summarizing conversations within sessions (reducing context length and storing results in PostgreSQL), and another for integrating and consolidating memory across sessions.
- Deployed LLM chatbots using CI/CD pipelines on GitHub Actions to automate deployment and improve the reliability of AI services.
- Developed a PII protection mechanism to mask sensitive entities before sending data to third-party LLM APIs.
- Created a personalized web-based chatbot that dynamically adapts to users through memory, interaction history, and preference tracking.
- Researched open-source embedding models as alternatives to OpenAI embeddings, reducing embedding-related costs by 15%.
- Built agentic AI in a TypeScript backend to manage chatbot memory, user interactions, and scheduling.
- Investigated fake image detection using proprietary LLM models to expand consultation service offerings.
  Technology Used: Typescript, Node js, Express JS, Milvus, FastAPI, github, docker, LLM, OpenAI, Groq, Amazon Bedrock, Azure AI, Presidio, ollama, llamacpp, PostGreSQL, DALL-E

Senior Machine Learning Engineer/ Data Scientist — June 2021 – June 2024
GDP Labs — *South Jakarta, Jakarta*

- Led a team of three engineers in building and designing on-premise RAG-based chatbot pipelines for internal client use cases, leveraging LangChain, LlamaIndex, ChromaDB embeddings, Flask, and prompt engineering.
- Implemented Named Entity Recognition (NER) to anonymize and de-anonymize entities in LLM-based chatbots before sending data to OpenAI, addressing security concerns for banking clients.
- Developed a chatbot powered by OpenAI LLMs on top of client data, improving data analysis efficiency by 25%.
- Researched and tested AI observability tools (e.g., DeepEval) to evaluate and enhance the reliability and performance of LLM applications.
- Fine-tuned Llama2 and Mistral models for one of the largest banks in Indonesia, achieving a 60% improvement in initial project outcomes.
- Advised clients on AWS for data analytics use cases, enabling automated data migration and optimizing efficiency by 30%.
- Built an MLOps system using AWS services (API Gateway, CloudWatch) to streamline data acquisition and model monitoring.
- Fine-tuned open-source multilingual embedding models with Q&A datasets, increasing retrieval performance by 15%.
- Automated evaluation of retrieval systems, accelerating end-to-end experimentation by 20%.
- Built inference scripts to deploy open-source models using Text Generation Inference, improving project performance by 15%.
- Experimented with prompt engineering techniques—including few-shot learning and chain-of-thought reasoning—to improve response quality.
- Optimized a coffee ingredient prediction system using LightGBM, reducing ingredient waste by 10%
  Technology Used: LangChain, LlamaIndex, Flask, ChromaDB, Elasticsearch, Prompt Engineering, AWS Cloud, GitHub, SQL, Pandas, Docker, Jupyter Notebook, Pytest, Boto3, Metabase, Kubernetes, LLM, OpenAI, PEFT, LoRA, guardrails, RAG, NER, Transformers, Grafana

Machine Learning Engineer Aug. 2019 – Jan. 2021
PT. Sugihart Digital Imaji *Duren Sawit, Jakarta*

- **People Density Heatmap**
  Developed people crowd density system and visualize it through heatmap to gain insight which area is the most crowded on specific time.
- **Face Recognition Attendance System**
  Built real-time face recognition for attendance system monitoring. The system speed up the attendance checking time with less than 2 seconds with accuracy more than 97%.
  Technology Used: OpenCV, Tensorflow, Pytorch, Darknet, Darkflow, Yolo, TensorRT, Deepstream, Jupyter Notebook, MySQL, netron, ONNX, threading

Data Scientist and Freelance projects Oct 2017 – Feb 2021
Various companies *Indonesia*

- Built text machine learning model for news filtering and achieve up to 95% accuracy.
- Clustered data for campaign needs and visualize it based on region by density, salary and religion.
- Researched and deployed NLP transformer model for Nanyang Technological University Singapore to facilitate their online course recommendation.
  Technology Used: AWS Cloud, github, pandas, docker, spacy, boto3, SentenceTransformer

Technical Skills

Programming Languages: Python, Javascript, SQL
Cloud Technologies: AWS (S3, Glue, Athena, EC2, Quicksight, Sagemaker, RDS, DMS, API Gateway, Lambda)
Machine Learning: Scikit-learn, PySpark, Tensorflow, Keras, FBProphet, PyOD, MLFlow
Big Data: Apache Spark, PrestoDB, TrinoDB

Open Source Contributor

JaidedAI/EasyOCR: Optical Character Recognition library that support 70+ languages

Certificate

AWS Certified Machine Learning Engineer – Associate | Oct 2025 Amazon Web Service

- Learn about AWS service for machine learning, deep learning and MLOps

Computer Vision Nanodegree | July 2020 Udacity

- Learn about facial keypoints, image captioning, and SLAM algorithm

Achievements

Mentor | July 2023 Garuda Hacks

- Mentor of 52 teams for Indonesia's Premier Hackathon

Advisor | June. 2023 Bangkit bootcamp by Google, GoTo, Traveloka

- Advisor to the 3 teams who built machine learning app using Google Cloud.

Best Employee | August. 2022 GDP Labs

- Best employee of the month out of 70+ employees

Top 10 | Nov. 2020 National Data Science Competition (NDSC) by Shopee

- Top 10 for Product Matching task (given image and text) in Advanced Category competition out of 139 teams with team name Babay